Shriram Pai

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CAREER OBJECTIVE

Electronics and Communication graduate with strong expertise in Embedded C, microcontrollers (PIC18F4580), and communication protocols (CAN, UART, SPI, I2C). Familiar with RTOS concepts including task scheduling and context switching. Seeking an entry-level Embedded/Firmware Engineer role to design and develop real-time embedded systems.

EDUCATION

NMAMIT, Nitte, Udupi **2021 – 2025**
Bachelor of Engineering in Electronics and Communication Engineering

Excellent Pu College, Kundapur **2019 – 2021**
Class 12th – Pre-University Course (PUC) *Grade: 98.14%*

New English School, Honnavar **2019**
Class 10th – Karnataka Secondary Education Examination Board (KSEEB) *Grade: 95.56%*

SKILLS

- **Languages:** C (Strong), C++
- **Embedded Systems:** Embedded C, PIC18F4580, 8051
- **Protocols:** UART, I2C, SPI, CAN
- **RTOS:** Task Scheduling, Context Switching, Basics of FreeRTOS
- **Operating Systems:** Linux, Windows
- **Tools:** MPLAB, Keil µVision
- **Concepts**: Data Structures, Linux System Programming, Interrupts, Memory Management

EXPERIENCE

Embedded Systems Training – Emertxe Information Technologies, Bangalore

- Completed intensive hands-on training in Embedded C, Linux System Programming, and Microcontroller-based firmware development.
- Created multiple system-level and embedded projects involving CAN, UDP sockets, and process management.

PROJECTS

CAN-Based Automotive Dashboard (using PIC and Embedded C)
- Developed a real-time automotive dashboard using PIC18F4580 microcontroller.
- Developed CAN protocol for ECU communication enabling real-time data exchange.
- Demonstrated understanding of real-time embedded systems and firmware development.

RTOS-Based Task Scheduling Simulation
- Implemented basic RTOS concepts such as task creation, scheduling, and context switching.
- Simulated multitasking behavior using priority-based execution.
- Strengthened understanding of real-time system design principles.

TFTP Client-Server Implementation (C, Linux, UDP Sockets)
- Developed a TFTP client-server application in C using UDP sockets with RRQ/WRQ handling, block sequencing, and retransmission logic.
- Gained hands-on experience in UDP socket programming, protocol design, and implementing reliability over an unreliable network.

MiniShell (Linux System Programming in C)
- Designed and implemented a custom Linux shell using C with support for command parsing and execution.
- Deep understanding of Linux process lifecycle (fork–exec–wait model). Improved knowledge of how operating systems manage processes internally.

Steganography using BMP Image (C Language)
- Developed and enforced a custom Linux shell using C with support for command parsing and execution.
- Strengthened understanding of bitwise operations, binary file handling, and low-level data manipulation in C.